$m, n \in \mathbb{N}_0$

$$\lim_{x\to 0} \frac{\sqrt[3]{1+mx+2x^2} - \sqrt[3]{1+x+x^2}}{x^n}$$

$$= \lim_{x\to 0} \frac{1+mx+2x^2-1-x-x^2}{x^n\left(\sqrt[3]{1+mx+2x^2}^2 + \sqrt[3]{1+mx+2x^2}\sqrt[3]{1+x+x^2} + \sqrt[3]{1+x+x^2}^2\right)}$$

$$= \lim_{x\to 0} \frac{x^2+(m-1)x}{x^n\left(\sqrt[3]{1+mx+2x^2}^2 + \sqrt[3]{1+mx+2x^2}\sqrt[3]{1+x+x^2} + \sqrt[3]{1+x+x^2}^2\right)}$$

$$= \lim_{x\to 0} \frac{x^2+(m-1)x}{x^n Z(x)}$$

kde

$$Z(x) = \sqrt[3]{1+mx+2x^2}^2 + \sqrt[3]{1+mx+2x^2}\sqrt[3]{1+x+x^2} + \sqrt[3]{1+x+x^2}^2$$

a jelikož je funkce spojitá v 0

$$\lim_{x\to 0} Z(x) = \lim_{x\to 0} \sqrt[3]{1+mx+2x^2}^2 + \sqrt[3]{1+mx+2x^2}\sqrt[3]{1+x+x^2} + \sqrt[3]{1+x+x^2}^2 = 3$$

a tedy zpět k původní limitě

$$= \frac{1}{3}\lim_{x\to 0} \frac{x^2+(m-1)x}{x^n} = \frac{1}{3}\lim_{x\to 0} \frac{x+m-1}{x^{n-1}}$$

Vyšetřím několik případů:

1. $m = 1$:
   1. $n = 0$

      $$\frac{1}{3}\lim_{x\to 0} \frac{x}{x^{-1}} = \frac{1}{3}\lim_{x\to 0} x^2 = 0$$

   2. $n = 1$

      $$\frac{1}{3}\lim_{x\to 0} \frac{x}{x^0} = \frac{1}{3}\lim_{x\to 0} x = 0$$

   3. $n = 2$

      $$\frac{1}{3}\lim_{x\to 0} \frac{x}{x^1} = \frac{1}{3}\lim_{x\to 0} \frac{x}{x} = \frac{1}{3}$$

   4. $n > 2$
      1. $2 \mid n$

         $$\frac{1}{3}\lim_{x\to 0} \frac{1}{x^{n-2}} = \infty$$

      2. $2 \nmid n$

         $\frac{1}{3}\lim_{x\to 0} \frac{1}{x^{n-2}}$ neex. - jednostranné limity se neshodují

1. $m \neq 1$:
   1. $n = 0$

$$\frac{1}{3} \lim_{x \to 0} \frac{x + m - 1}{x^{-1}} = \frac{1}{3} \lim_{x \to 0} x^2 + mx - x = 0$$

   2. $n = 1$

$$\frac{1}{3} \lim_{x \to 0} \frac{x + m - 1}{x^0} = \frac{1}{3} \lim_{x \to 0} x + m - 1 = \frac{m - 1}{3}$$

   3. $n > 1$
      1. $2 \nmid n \wedge m = 0$

$$\frac{1}{3} \lim_{x \to 0} \frac{x - 1}{x^{n-1}} = -\infty$$

      2. $2 \nmid n \wedge m > 1$

$$\frac{1}{3} \lim_{x \to 0} \frac{x + m - 1}{x^{n-1}} = \infty$$

      3. $2 \mid n$

$$\frac{1}{3} \lim_{x \to 0} \frac{x + m - 1}{x^{n-1}} \text{ neex. - jednostranné limity se neshodují}$$